LDA Homework 1 (2025-2026)

Question 1

Check if Data is Balanced

From the description of the experiment we note that there is dropout during follow up. Before using any summary methods to check the mean, variancce and correlation structure we first check if the data is balanced. We see that there is missingness in the data as shown in figure 1 below. Patients with higher baseline BPRS scores tend to miss more follow-up visits. There is also an increase of of the percentage of missingness over time, reaching above 40% by year 6. Because simple by time averages overweight those who remain and underrepresent those who dropped out (patients with higher baseline BPRS scores), they can paint an a biased trend if simple bprs by time averages are used. This pattern suggests data are not missing completely at random, so we use smoothing methods that are better suited for unbalanced data

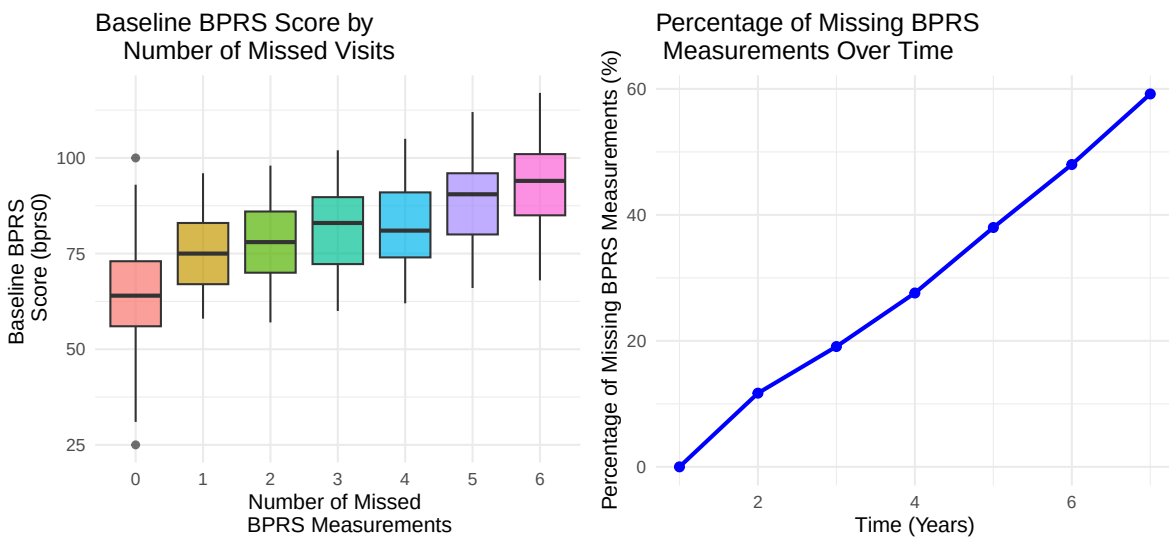


Figure 1: Baseline BPRS by number of missed visits (left) and percentage missing over time (right).

Zero-variance check for baseline Tau PET (taupet0)

From the table, taupet0 still varies through Year 2 (SD drops from 0.116 to 0.023), but from Year 3 onwards it has zero variance (only one unique value among those with BPRS observed). That means taupet0 cannot explain any between-patient differences after Year 2 and coefficients for those years are not estimable. We should not use taupet0 from year 3 as this variable would not provide information on growth trajectories from here onwards.

Table 1: Baseline taupet0 variability among patients with observed BPRS at each year

n_patients	n_unique_taupet0	n_unique_abet0	sd_taupet0	sd_abet0	year	zero_variance
1253	27	67	0.1159765	0.4535830	0	FALSE
1106	13	65	0.0492840	0.4208376	1	FALSE
1014	6	62	0.0231188	0.3967379	2	FALSE
907	1	55	0.0000000	0.3730319	3	TRUE
777	1	50	0.0000000	0.3245475	4	TRUE
652	1	39	0.0000000	0.2718052	5	TRUE
511	1	22	0.0000000	0.2131188	6	TRUE

Average loess smoothing

Categorical Variables

To explore potential subgroup differences, we create LOESS smoothed plots of BPRS over time stratified by categorical variables: sex, education, job status, living situation, and clinical center. Across these variables, BPRS increases over time. Most subgroup curves are roughly parallel, showing main effects without interactions for variables like sex, job status, and living situation. The four education categories overlap almost completely and there is no difference in trend. This means there is unlikely a significant effect of education on BPRS trajectory. Taupet also shows distinct level shifts with roughly parallel trends, indicating a main effect without interaction. was a variable with only one measurement where it was not missing hence a zero variance variable.

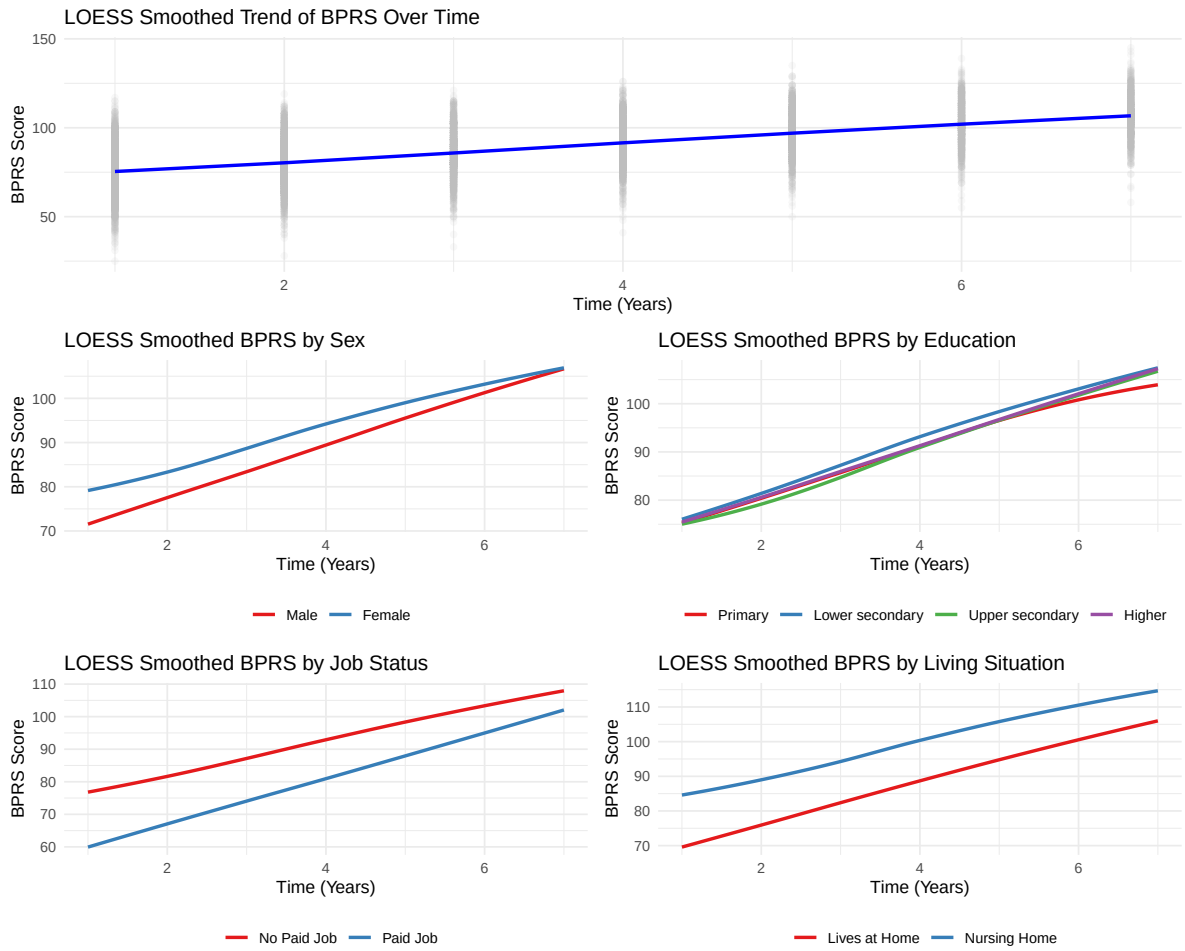


Figure 2: LOESS Smoothed BPRS Over Time by Subgroups.
Overall average (top), sex, education, job status,
living situation, and clinical center.

Continuous Variables

To explore continuous predictors, we plotted LOESS-smoothed BPRS against age, BMI, baseline BPRS (bprs0), and baseline CDR-SB (cdrsb0), using separate curves for each follow-up year. Higher age and higher baseline BPRS are both consistently associated with higher BPRS at every visit. BMI shows a curved pattern that peaks in the mid-20s, suggesting that a non-linear term would describe it better than a straight line. Baseline CDR-SB shows only a weak and slightly curved association with BPRS.

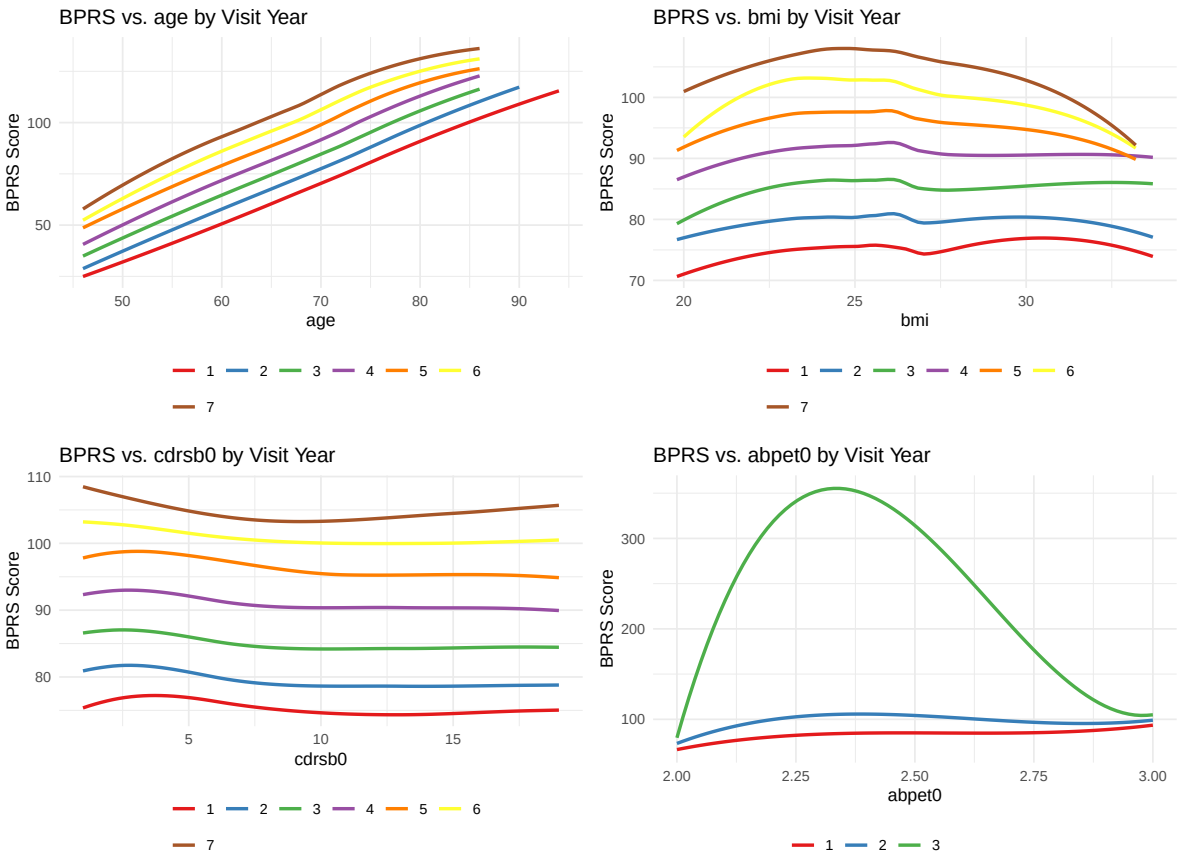


Figure 3: LOESS Smoothed BPRS Over Continuous Predictors by Visit Year.
Age, BMI, baseline BPRS (bprs0), and baseline CDR−SB (cdrsb0).

Trial Centers Plot

To check whether clinical center (trial) affects the mean structure, we plot(figure 4) LOESS-smoothed BPRS over time separately by trial centre. We see clear separate curves between centers, indicating that trial should be included as a random effect in the model to account for between center variability.

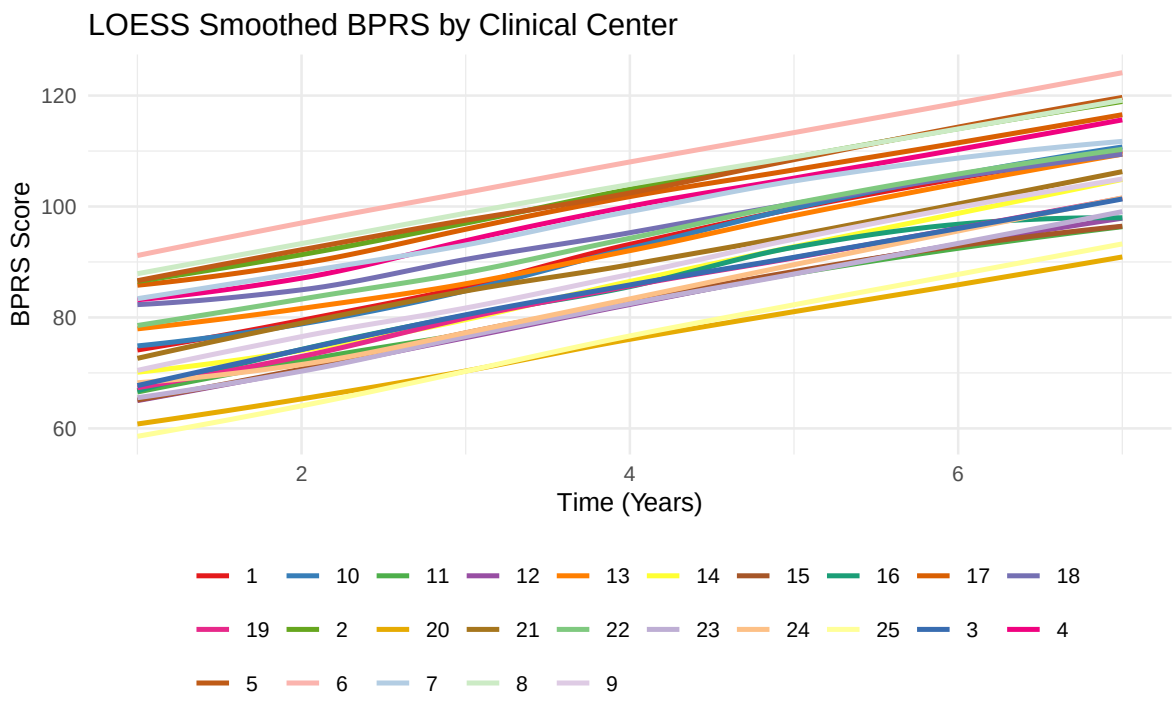


Figure 4: LOESS Smoothed BPRS Over Time by Clinical Center.

Variance & Correlation Structure

The smoothed squared residuals shows a flat residual curve, we dont observe a heteroscedastic pattern, meaning the spread of patients BPR's readings tranjectory around there expected readings doesn't change a lot. The semi-variogram also rises with the time gap between measurements, showing that observations from the same patient are more similar when close together and become less similar as the gap widens. These patterns indicate increasing variance over time and a clear within-patient correlation that weakens with distance in time. Models therefore need to allow for changing variances and a correlation structure that decays over time. The plots are shown in figure 5 below.

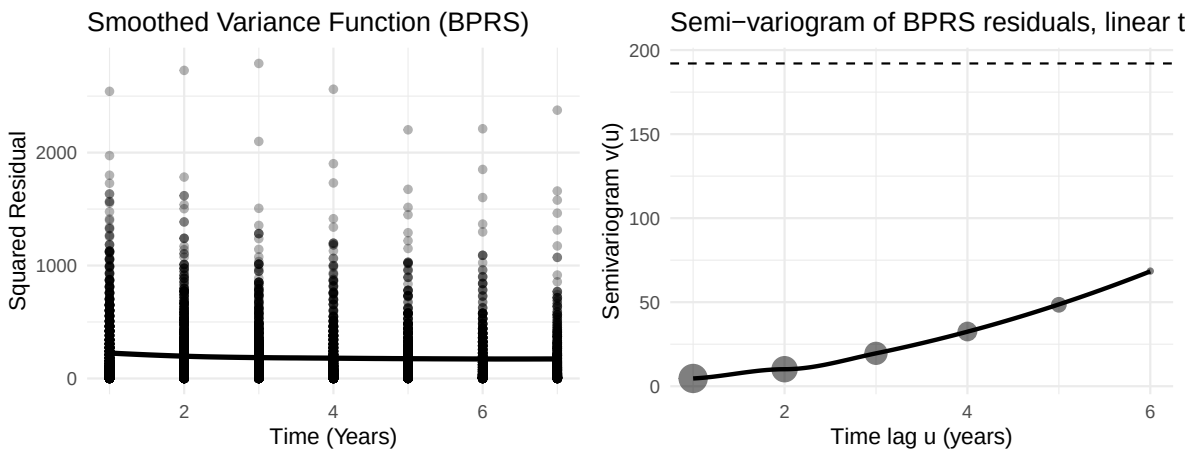


Figure 5: Smoothed Variance Function (left) and Semi−variogram (right) of BPRS Residuals.

Question 1: Conclusions and modeling implications

- Mean structure: BPRS rises steadily over time. Subgroup curves are mostly parallel with level shifts by sex, job, living situation, and clear vertical shifts by clinical center; education shows little separation.
- Variance structure: The smoothed squared-residual function is roughly flat across visits (no visual heteroscedasticity observed). We will fit a model with un assumed heterogeneous variance and compare this with a constant variance model using AIC.
- Correlation structure: The semivariogram increases with time lag, indicating strong within-patient correlation at short lags that decays as visits are farther apart.
- Missingness: Dropout increases over time and is more common for patients with higher baseline BPRS, so simple time-point averages would be biased
- Modeling implications: For population-average inference, use repeated measures with with unstructured covariance structure and compare with AR(1) correlation
- For subject-specific inference, use a mixed model with patient random intercept and slope; include a random intercept for clinical center, because there is a clear clinical site differences

- we don't recommend using taupet 0 as a covariate in the model as it has zero variance from year 3 onwards, for the participants who's BPRS measurement were taken at year 3 and beyond. It will not contribute to the model

R for mixed models

For all analysis and and especially repeated measures models we fit the models in R using the nlme package for population-average analyses. For these models we used `nlme::gls`a repeated-measures approach where we specify the within-patient covariance (e.g., unstructured with visit-specific variances or AR(1) with heterogeneity). This mirrors the method used in SAS (PROC MIXED with a REPEATED statement): it estimates average trajectories while properly accounting for the correlation between a patient's repeated observations, without introducing individual random effects. For Question 5, where objective is to study patient-specific trajectories, we used `lme4::lmer` to fit a mixed model with random intercepts and random slopes, allowing each patient to have their own baseline level and rate of change while still estimating the overall population effects.

Question 2

Individual profiles & per-patient slopes

To visualise heterogeneity in longitudinal evolution, BPRS trajectories of 40 random patients were plotted. These individual profiles show variation in baseline symptom severity, together with increasing trends over time. The profiles also illustrate the unbalanced nature of the data, as subjects contribute different numbers of observations. These observations help justify the choice of summary statistics for longitudinal data.

Figure 6: Individual BPRS Profiles (left) and Distribution of Per−Patient Slopes (right).

Change Score Analysis

The change score is defined as the difference between the last observed BPRS score and the baseline BPRS score for each patient. A linear model is fitted to determine which baseline covariates predict this change. The model is:

$$\begin{aligned}\text{ChangeScore}_i =& \beta_0 + \beta_1 \cdot \text{sex}_i + \beta_2 \cdot \text{age}_i + \beta_3 \cdot \text{bmi}_i + \beta_4 \cdot \text{job}_i + \beta_5 \cdot \text{adl}_i \\ &+ \beta_6 \cdot \text{wzc}_i + \beta_7 \cdot \text{cdrsb0}_i + \beta_8 \cdot \text{abpet0}_i + \beta_9 \cdot \text{taupet0}_i + \epsilon_i\end{aligned}$$

where:

- $\text{ChangeScore}_i = \text{BPRS}_{\text{last},i} - \text{BPRS}_{0,i}$ for patient i.
- β_0 is the intercept: expected change for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates equal to 0.
- β_1 is the effect of sex on change: Female vs Male.
- β_2 is the effect of age on change: per 1-year increase.
- β_3 is the effect of BMI on change: per 1-unit increase.
- β_4 is the effect of job on change: Paid Job vs No Paid Job.
- β_5 is the effect of ADL on change: per 1-unit increase.
- β_6 is the effect of living situation (wzc) on change: Nursing Home vs Lives at Home.
- β_7 is the effect of baseline CDR-SB (cdrsb0) on change: per 1-unit increase.
- β_8 is the effect of baseline Amyloid PET (abpet0) on change: per 1-unit increase.
- β_9 is the effect of baseline Tau PET (taupet0) on change: per 1-unit increase.
- ϵ_i is the error term.

Table 2: Change Score Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	7.35 [-8.54, 23.23]	8.10	0.91	0.36
sexFemale	-0.81 [-1.84, 0.22]	0.52	-1.55	0.12
age	0.26 [0.10, 0.43]	0.08	3.15	<0.01
bmi	-0.02 [-0.26, 0.21]	0.12	-0.19	0.85
jobPaid Job	-29.64 [-33.02, -26.27]	1.72	-17.22	<0.01
adl	4.96 [4.54, 5.39]	0.22	22.95	<0.01
wzcNursing Home	-6.86 [-8.00, -5.72]	0.58	-11.83	<0.01
cdrsb0	-0.16 [-0.22, -0.09]	0.03	-4.55	<0.01
abpet0	-6.61 [-8.42, -4.81]	0.92	-7.18	<0.01
taupet0	-5.19 [-9.65, -0.73]	2.27	-2.28	0.02

Separate analyses for each follow-up year

For each follow-up year $t \in \{1, \dots, 6\}$, we fit an ordinary linear model relating the BPRS score at year t to baseline covariates:

$$\begin{aligned} extBPRS_{it} = \alpha_{0t} &+ \alpha_{1t} \cdot \text{sex}_i + \alpha_{2t} \cdot \text{age}_i + \alpha_{3t} \cdot \text{bmi}_i + \alpha_{4t} \cdot \text{job}_i + \alpha_{5t} \cdot \text{adl}_i \\ &+ \alpha_{6t} \cdot \text{wzc}_i + \alpha_{7t} \cdot \text{cdrsb0}_i + \alpha_{8t} \cdot \text{abpet0}_i + \alpha_{9t} \cdot \text{taupet0}_i + \varepsilon_{it}. \end{aligned}$$

where, for each fixed year t:

- α_{0t} is the expected BPRS at year t for the reference patient (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- α_{1t} is the difference at year t for Female vs Male.
- α_{2t} is the change at year t per 1-year increase in age.
- α_{3t} is the change at year t per 1-unit increase in BMI.
- α_{4t} is the difference at year t for Paid Job vs No Paid Job.
- α_{5t} is the change at year t per 1-unit increase in ADL.
- α_{6t} is the difference at year t for Nursing Home vs Lives at Home.
- α_{7t} is the change at year t per 1-unit increase in baseline CDR-SB (cdrsb0).
- α_{8t} is the change at year t per 1-unit increase in baseline Amyloid PET (abpet0).
- α_{9t} is the change at year t per 1-unit increase in baseline Tau PET (taupet0).
- ε_{it} is the error term at year t.

Table 3: Separate Linear Models for Each Follow-up Year

covariate	Year0	Year1	Year2	Year3	Year4	Year5	Year6
(Intercept)	-122.72 [-126.28, -119.16]	-119.78 [-126.03, -113.52]	-105.57 [-120.15, -90.99]	-104.92 [-111.33, -98.51]	-101.46 [-109.63, -93.29]	-95.55 [-106.07, -85.02]	-87.73 [-101.67, -73.79]
abpet0	0.34 [-0.07, 0.74]	0.20 [-0.29, 0.69]	0.03 [-0.61, 0.68]	-0.13 [-0.98, 0.71]	-0.03 [-1.18, 1.13]	-0.10 [-1.74, 1.54]	-1.11 [-3.75, 1.54]
adl	2.01 [1.92, 2.11]	2.17 [2.05, 2.29]	2.25 [2.09, 2.41]	2.11 [1.91, 2.32]	2.26 [2.00, 2.52]	2.37 [2.02, 2.71]	2.42 [1.96, 2.87]
age	2.36 [2.33, 2.40]	2.41 [2.36, 2.45]	2.42 [2.37, 2.48]	2.41 [2.34, 2.48]	2.45 [2.36, 2.54]	2.48 [2.37, 2.58]	2.50 [2.37, 2.64]
bmi	0.49 [0.44, 0.55]	0.52 [0.45, 0.58]	0.51 [0.42, 0.59]	0.54 [0.43, 0.65]	0.56 [0.42, 0.70]	0.51 [0.33, 0.69]	0.47 [0.23, 0.71]
cdrsb0	-0.02 [-0.04, -0.01]	-0.04 [-0.06, -0.02]	-0.06 [-0.08, -0.04]	-0.08 [-0.11, -0.05]	-0.13 [-0.17, -0.09]	-0.15 [-0.20, -0.10]	-0.17 [-0.23, -0.10]
jobPaid Job	-18.33 [-19.09, -17.57]	-19.24 [-20.17, -18.30]	-19.77 [-20.97, -18.56]	-19.01 [-20.54, -17.47]	-20.19 [-22.13, -18.25]	-20.86 [-23.30, -18.41]	-21.07 [-24.18, -17.96]
sexFemale	-0.28 [-0.51, -0.05]	-0.40 [-0.68, -0.12]	-0.18 [-0.54, 0.18]	-0.22 [-0.68, 0.25]	-0.34 [-0.94, 0.26]	-0.37 [-1.15, 0.41]	-0.57 [-1.64, 0.49]
taupet0	0.43 [-0.57, 1.43]	0.37 [-2.34, 3.09]	-4.06 [-11.47, 3.35]	NA	NA	NA	NA
wzcNursing Home	1.69 [1.43, 1.94]	1.94 [1.63, 2.25]	2.10 [1.69, 2.51]	2.24 [1.69, 2.80]	1.94 [1.18, 2.71]	1.43 [0.34, 2.52]	2.16 [0.42, 3.90]

Question 3

Motivation Repeated-Measures Model

To estimate population effects, we use a repeated-measures multivariate model. This approach treats each patient's yearly BPRS readings as not independent. A normal multivariate linear model assumes all observations are independent, which is not realistic here because the same

patient is measured several times. The repeated-measures model correctly handles the correlation within a person, the uneven follow-up due to dropout, and the changing variability across visits.This model uses all available data and gives more reliable averages and standard errors. We start from a saturated repeated-measures model where time, all baseline characteristics, and their interactions with time are included. This gives every variable a way to influence both the starting BPRS level and the change over time, so that we do not miss important effects when we later simplify the model. For the repeated measurements within a patient, we allow a very flexible pattern for how BPRS scores at different visits are related to each other, without forcing a specific correlation pattern in advance. We do not use this saturated model to report final clinical results, we use it as a screening model and then simplify the mean structure. In question 2 we had yearly results but analysing each year separately would mean running several models for the same question/hypothesis. This increases the chance of getting a significant result just by chance (Type 1 error), because we are not using all observations together it also leads to lower power. Separate analyses also do not answer the main question of the study, which is how BPRS changes over time and how baseline factors relate to that change. They ignore that measurements from the same patient are correlated, which can give misleading results. The repeated-measures model avoids these problems by using all time points at once and by correctly accounting for the fact that each patient contributes several, correlated measurements.

$$
\begin{aligned}
E[Y_{ij}|X_i] = &\beta_0 + \beta_1 \cdot \text{time}_j + \beta_2 \cdot \text{edu}_i + \beta_3 \cdot \text{sex}_i + \beta_4 \cdot \text{age}_i + \beta_5 \cdot \text{bmi}_i + \beta_6 \cdot \text{job}_i \\
&+ \beta_7 \cdot \text{adl}_i + \beta_8 \cdot \text{wzc}_i + \beta_9 \cdot \text{cdrsb0}_i + \beta_{10} \cdot \text{abpet0}_i \\
&+ \beta_{11} \cdot (\text{time}_j \times \text{sex}_i) + \beta_{12} \cdot (\text{time}_j \times \text{age}_i) + \beta_{13} \cdot (\text{time}_j \times \text{bmi}_i) \\
&+ \beta_{14} \cdot (\text{time}_j \times \text{job}_i) + \beta_{15} \cdot (\text{time}_j \times \text{adl}_i) + \beta_{16} \cdot (\text{time}_j \times \text{wzc}_i) \\
&+ \beta_{17} \cdot (\text{time}_j \times \text{cdrsb0}_i) + \beta_{18} \cdot (\text{time}_j \times \text{abpet0}_i)
\end{aligned}
$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
- β_0 is the intercept: expected BPRS at time 0 for the reference patient (Male, Primary education, No Paid Job, Lives at Home), with continuous covariates at 0.
- β_1 is the main effect of time: average yearly change in BPRS for the reference patient.
- β_2 is the (vector of) main effect(s) of education: contrasts for Lower secondary, Upper secondary, and Higher vs Primary (written compactly as $\beta_2 \, \text{edu}_i$).
- β_3 is the main effect of sex: Female vs Male at time 0.
- β_4 is the main effect of age: per 1-year increase at baseline.
- β_5 is the main effect of BMI: per 1-unit increase at baseline.
- β_6 is the main effect of job: Paid Job vs No Paid Job at time 0.

- β_7 is the main effect of ADL: per 1-unit increase at baseline.
- β_8 is the main effect of living situation (wzc): Nursing Home vs Lives at Home at time 0.
- β_9 is the main effect of baseline CDR-SB (cdrsb0): per 1-unit increase at baseline.
- β_{10} is the main effect of baseline Amyloid PET (abpet0): per 1-unit increase at baseline.
- β_{11} is the time-by-sex interaction: additional yearly change for Female vs Male.
- β_{12} is the time-by-age interaction: modification of yearly change per 1-year increase in age.
- β_{13} is the time-by-BMI interaction: modification of yearly change per 1-unit increase in BMI.
- β_{14} is the time-by-job interaction: additional yearly change for Paid Job vs No Paid Job.
- β_{15} is the time-by-ADL interaction: modification of yearly change per 1-unit increase in ADL.
- β_{16} is the time-by-living situation (wzc) interaction: additional yearly change for Nursing Home vs Lives at Home.
- β_{17} is the time-by-cdrsb0 interaction: modification of yearly change per 1-unit increase in baseline CDR-SB.
- β_{18} is the time-by-abpet0 interaction: modification of yearly change per 1-unit increase in baseline Amyloid PET.

Notes: Categorical predictors (sex, edu, job, wzc) are encoded as factors in R. Their coefficients represent contrasts relative to the reference levels shown above; for education, the single symbol β_2 denotes a vector of three contrasts (levels 2–4 versus Primary).

The within-patient errors $e_i = (e_{i1}, \dots, e_{in_i})^T$ are assumed to follow a multivariate normal distribution, $e_i \sim N(0, \Sigma_i)$, where Σ_i is a covariance matrix that accounts for correlation and heteroscedasticity. In this model, we use an unstructured correlation matrix (`corSymm`) and allow for different variances at each time point (`varIdent`).

Note: taupet0 terms (main effect and time interaction) were intentionally excluded from this model due to near-zero variability at later follow-up waves (see earlier diagnostic section), which prevents stable estimation of their effects in a marginal mean model.

The complex repeated-measures model that includes time, all baseline characteristics, and all their interactions with time, while allowing for unstructured covariance structure. This complex model confirms that BPRS increases over time and that age, BMI, ADL, job status, and living in a nursing home are all related to BPRS levels. However, most of the time-by-covariate interactions are small and not clearly different from zero, which tells us we can safely move to a simpler mean structure without losing important clinical information.

Table 4: Gls Full Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	-129.58 [-133.00, -126.15]	1.75	-74.20	<0.01
time	5.96 [4.27, 7.64]	0.86	6.91	<0.01
eduLower secondary	0.20 [-0.13, 0.52]	0.17	1.19	0.23
eduUpper secondary	0.12 [-0.19, 0.42]	0.16	0.75	0.46
eduHigher	0.19 [-0.11, 0.50]	0.16	1.23	0.22
sexFemale	-0.24 [-0.49, 0.00]	0.13	-1.95	0.05
age	2.36 [2.32, 2.40]	0.02	126.37	<0.01
bmi	0.51 [0.45, 0.57]	0.03	17.00	<0.01
jobPaid Job	-18.13 [-18.91, -17.35]	0.40	-45.69	<0.01
adl	2.00 [1.90, 2.10]	0.05	38.57	<0.01
wzcNursing Home	1.63 [1.35, 1.92]	0.15	11.16	<0.01
cdrsb0	0.00 [-0.01, 0.02]	0.01	0.21	0.83
abpet0	0.34 [-0.10, 0.78]	0.22	1.52	0.13
time:sexFemale	-0.03 [-0.16, 0.09]	0.06	-0.52	0.61
time:age	0.01 [-0.00, 0.03]	0.01	1.59	0.11
time:bmi	-0.00 [-0.03, 0.03]	0.01	-0.06	0.95
time:jobPaid Job	-0.39 [-0.78, 0.01]	0.20	-1.90	0.06
time:adl	0.05 [-0.00, 0.11]	0.03	1.95	0.05
time:wzcNursing Home	0.14 [-0.00, 0.29]	0.07	1.90	0.06
time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.10	<0.01
time:abpet0	-0.07 [-0.29, 0.16]	0.11	-0.58	0.56

Simplified Mean Structure

For this step we fit a repeated-measures model with a simplified mean structure but the same flexible covariance as before. In this model, time and all main baseline characteristics except education remain in the mean structure, and only one time interaction (time with baseline CDR-SB) is kept. The estimated correlation matrix shows clear positive correlation between repeated BPRS scores within a patient, especially for visits that are closer in time, and the visit-specific standard deviations increase from baseline to later years, contradicting with our earlier variance plots showing homoscedastic variance. We will use this reduced-mean, flexible-covariance model as the starting point when we compare alternative, simpler covariance structures.

$$\begin{aligned} E[Y_{ij}|X_i] =& \beta_0 + \beta_1 \cdot \text{time}_j + \beta_2 \cdot \text{sex}_i + \beta_3 \cdot \text{age}_i + \beta_4 \cdot \text{bmi}_i + \beta_5 \cdot \text{job}_i \\ &+ \beta_6 \cdot \text{adl}_i + \beta_7 \cdot \text{wzc}_i + \beta_8 \cdot \text{cdrsb0}_i + \beta_9 \cdot \text{taupet0}_i + \beta_{10} \cdot \text{abpet0}_i \\ &+ \beta_{11} \cdot (\text{time}_j \times \text{cdrsb0}_i) \end{aligned}$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
- β_0 is the intercept: expected BPRS at time 0 for the reference patient (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- β_1 is the main effect of time: average yearly change in BPRS for the reference patient.
- β_2 is the main effect of sex: Female vs Male at time 0.
- β_3 is the main effect of age: per 1-year increase at baseline.
- β_4 is the main effect of BMI: per 1-unit increase at baseline.
- β_5 is the main effect of job: Paid Job vs No Paid Job at time 0.
- β_6 is the main effect of ADL: per 1-unit increase at baseline.
- β_7 is the main effect of living situation (wzc): Nursing Home vs Lives at Home at time 0.
- β_8 is the main effect of baseline CDR-SB (cdrsb0): per 1-unit increase at baseline.
- β_9 is the main effect of baseline Tau PET (taupet0): per 1-unit increase at baseline.
- β_{10} is the main effect of baseline Amyloid PET (abpet0): per 1-unit increase at baseline.
- β_{11} is the time-by-cdrsb0 interaction: modification of yearly change per 1-unit increase in baseline CDR-SB.

The covariance structure remains the same as in the full model.

Table 5: Simplfied mean structiure Unstructured Covariance Results

term	estimate	std.error	statistic	p.value
(Intercept)	-131.59 [-134.11, -129.08]	1.28	-102.43	<0.01
time	7.22 [7.14, 7.30]	0.04	179.38	<0.01
sexFemale	-0.27 [-0.46, -0.07]	0.10	-2.72	<0.01
age	2.39 [2.37, 2.42]	0.01	212.64	<0.01
bmi	0.51 [0.47, 0.55]	0.02	22.80	<0.01
jobPaid Job	-18.69 [-19.30, -18.08]	0.31	-59.80	<0.01
adl	2.08 [2.01, 2.16]	0.04	53.17	<0.01
wzcNursing Home	1.84 [1.62, 2.05]	0.11	16.86	<0.01
cdrsb0	0.00 [-0.01, 0.02]	0.01	0.35	0.73
time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.29	<0.01

LR test between full and reduced mean structure models

The likelihood ratio test compares the complex mean structure model (with all time–covariate interactions) to the reduced model (only time $\times$ CDR-SB). The test statistic is 13.78 with a p-value of 0.25, which means there is no evidence that the extra interaction terms improve the fit. This shows the simpler mean structure explains the data just as well as the saturated one, while using fewer parameters, so we keep the reduced model as our working mean structure.

Table 6: Likelihood Ratio Test between Full and Reduced Mean Structure Models

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
gls_full_mean_structure	1	49	27318.29	27648.33	-13610.14		NA	NA
gls_reduced_mean_structure	2	38	27310.07	27566.02	-13617.03	1 vs 2	13.78	0.25

Reduced Covariance Structure

For this step, we keep the same simplified mean structure and only change how we model the correlation over time. The aim is to check whether a simpler covariance structure is sufficient. We fit a repeated-measures model with an AR(1) correlation, where each visit is mostly correlated with the previous one and the correlation weakens as the time gap increases. We still allow variability to differ by visit. Clinically, this means measurements taken close in time tend to be similar, and later visits show more spread, but the structure is quite restricted.

When we compare this AR(1) model to the more flexible model with a fully general correlation matrix, the likelihood ratio test shows (L.Ratio ≈ 1233, $p < 0.0001$) and AIC/BIC worsen for AR(1). This indicates that the data pattern is too complex to be captured by this simpler

correlation structure, so for population-average inference we keep the more flexible covariance from the previous model.

Table 7: Marginal Model with AR(1) Covariance Results

term	estimate	std.error	statistic	p.value
(Intercept)	-131.98 [-135.20, -128.75]	1.64	-80.27	<0.01
time	7.24 [7.17, 7.31]	0.03	208.57	<0.01
sexFemale	-0.28 [-0.53, -0.03]	0.13	-2.24	0.03
age	2.40 [2.37, 2.43]	0.01	166.48	<0.01
bmi	0.50 [0.45, 0.56]	0.03	17.57	<0.01
jobPaid Job	-18.91 [-19.69, -18.12]	0.40	-47.07	<0.01
adl	2.10 [2.00, 2.20]	0.05	41.83	<0.01
wzcNursing Home	1.84 [1.57, 2.12]	0.14	13.25	<0.01
cdrsb0	0.00 [-0.02, 0.03]	0.01	0.33	0.74
time:cdrsb0	-0.02 [-0.03, -0.02]	0.00	-6.67	<0.01

Reduced mean structure vs reduced covariance structure

Table 8: Likelihood Ratio Test between Reduced Mean Structure and Reduced Covariance Structure Models

	Model	df	AIC	BIC	logLik	Test	L.Ratio	p-value
gls_reduced_mean_structure	1	38	27310.07	27566.02	-13617.03		NA	NA
gls_reduced_cov_structure	2	18	28503.30	28624.54	-14233.65	1 vs 2	1233.23	0

Discussion of Question 3 Results

For Question 3, our final repeated-measures model uses a reduced mean structure with a flexible covariance. On the mean side, BPRS increases on average by about 7 points per year. After adjustment, higher age, higher BMI, worse ADL, having no paid job, and living in a nursing home are all associated with higher BPRS levels. Women have slightly lower BPRS than men. Baseline CDR-SB does not change the level much, but the time×CDR-SB term is clearly negative, meaning that patients who start with better cognition tend to show a faster increase in BPRS, while those starting with worse cognition worsen more slowly.

The covariance part of the model allows each visit to have its own variability and lets all time points within a patient be correlated without imposing a simple pattern. This matches what we saw in the exploratory plots: BPRS measurements from the same patient are strongly related, especially at close time points, and the spread increases over time. By using this flexible structure, the model can give realistic standard errors for the population averages.

This marginal model has some limitations compared with a mixed-effects model. First, it only gives population-average effects; it does not provide patient-specific trajectories. Second, differences between clinical centres (trial) are not modelled. The LOESS plot by centre shows centres with parallel but clearly shifted curves, suggesting centre-level variability in BPRS levels over time. In the current model, this extra variation is absorbed into the residual covariance, rather than being captured as a separate trial effect. In Question 5, a mixed-effects model with random intercepts (and slopes) for patients, and a random intercept for trial, will better describe these centre and patient differences and therefore provide a richer description of individual and centre-specific intercepts.

Question 4

In the first stage of a two-stage analysis, we fit a separate linear regression model for each patient to estimate their individual intercept and slope over time. For each patient i, the model is:

$$Y_{ij} = \beta_{0i} + \beta_{1i} \cdot \text{time}_j + e_{ij}$$

where:

- Y_{ij} is the BPRS score for patient i at time j.
- β_{0i} is the patient-specific intercept (estimated baseline BPRS).
- β_{1i} is the patient-specific slope (estimated rate of change in BPRS).
- e_{ij} are the individual error terms, assumed to be independent with mean zero.

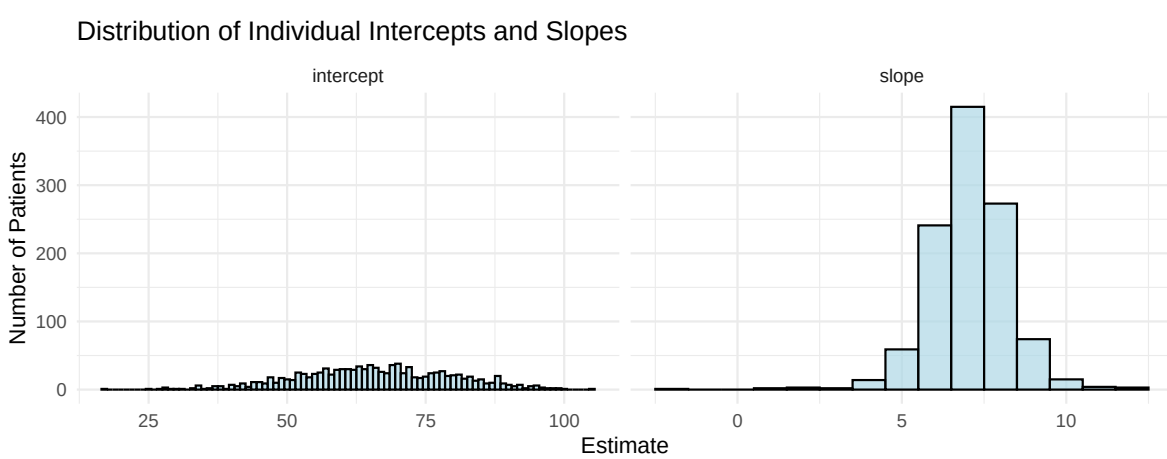


Figure 7: Individual BPRS Profiles (left) and
Distribution of Individual Intercepts and Slopes (right).

Stage 2: Relate Intercepts and Slopes to Covariates

Model for the Intercepts

In the second stage, we model the estimated patient-specific intercepts ($\hat{\beta}_{0i}$) as a function of baseline covariates to understand which factors predict the initial BPRS score. The model is:

$$\begin{aligned}\hat{\beta}_{0i} =& \gamma_{00} + \gamma_{01} \cdot \text{sex}_i + \gamma_{02} \cdot \text{age}_i + \gamma_{03} \cdot \text{bmi}_i + \gamma_{04} \cdot \text{job}_i + \gamma_{05} \cdot \text{adl}_i \\ &+ \gamma_{06} \cdot \text{wzc}_i + \gamma_{07} \cdot \text{cdrsb0}_i + \gamma_{08} \cdot \text{abpet0}_i + \gamma_{09} \cdot \text{taupet0}_i + \epsilon_{0i}\end{aligned}$$

where:

- $\hat{\beta}_{0i}$ is the estimated intercept for patient i from Stage 1.
- γ_{00} is the average intercept for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- γ_{01} is the intercept difference for Female vs Male.
- γ_{02} is the intercept change per 1-year increase in age.
- γ_{03} is the intercept change per 1-unit increase in BMI.
- γ_{04} is the intercept difference for Paid Job vs No Paid Job.
- γ_{05} is the intercept change per 1-unit increase in ADL.
- γ_{06} is the intercept difference for Nursing Home vs Lives at Home.
- γ_{07} is the intercept change per 1-unit increase in baseline CDR-SB (cdrsb0).
- γ_{08} is the intercept change per 1-unit increase in baseline Amyloid PET (abpet0).
- γ_{09} is the intercept change per 1-unit increase in baseline Tau PET (taupet0).
- ϵ_{0i} is the error term.

Model for the Slopes

Similarly, we model the estimated patient-specific slopes ($\hat{\beta}_{1i}$) as a function of baseline covariates to see what predicts the rate of change in BPRS. The model is:

$$\begin{aligned}\hat{\beta}_{1i} =& \gamma_{10} + \gamma_{11} \cdot \text{sex}_i + \gamma_{12} \cdot \text{age}_i + \gamma_{13} \cdot \text{bmi}_i + \gamma_{14} \cdot \text{job}_i + \gamma_{15} \cdot \text{adl}_i \\ &+ \gamma_{16} \cdot \text{wzc}_i + \gamma_{17} \cdot \text{cdrsb0}_i + \gamma_{18} \cdot \text{abpet0}_i + \gamma_{19} \cdot \text{taupet0}_i + \epsilon_{1i}\end{aligned}$$

where:

- $\hat{\beta}_{1i}$ is the estimated slope for patient i from Stage 1.

Table 9: Two-Stage Intercept Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	-130.85 [-137.31, -124.39]	3.29	-39.76	<0.01
sexFemale	-0.22 [-0.51, 0.06]	0.15	-1.52	0.13
age	2.36 [2.32, 2.41]	0.02	101.88	<0.01
bmi	0.51 [0.45, 0.58]	0.03	15.07	<0.01
jobPaid Job	-18.28 [-19.25, -17.32]	0.49	-37.15	<0.01
adl	2.03 [1.90, 2.15]	0.06	31.98	<0.01
wzcNursing Home	1.54 [1.22, 1.86]	0.16	9.41	<0.01
cdrsb0	0.00 [-0.02, 0.02]	0.01	0.12	0.90
abpet0	0.35 [-0.16, 0.86]	0.26	1.35	0.18
taupet0	0.53 [-2.27, 3.33]	1.43	0.37	0.71

- γ_{10} is the average slope for the reference individual (Male, No Paid Job, Lives at Home), with continuous covariates at 0.
- γ_{11} is the slope difference for Female vs Male.
- γ_{12} is the slope change per 1-year increase in age.
- γ_{13} is the slope change per 1-unit increase in BMI.
- γ_{14} is the slope difference for Paid Job vs No Paid Job.
- γ_{15} is the slope change per 1-unit increase in ADL.
- γ_{16} is the slope difference for Nursing Home vs Lives at Home.
- γ_{17} is the slope change per 1-unit increase in baseline CDR-SB (cdrsb0).
- γ_{18} is the slope change per 1-unit increase in baseline Amyloid PET (abpet0).
- γ_{19} is the slope change per 1-unit increase in baseline Tau PET (taupet0).
- ϵ_{1i} is the error term.

Question 5

Variance components from mixed model

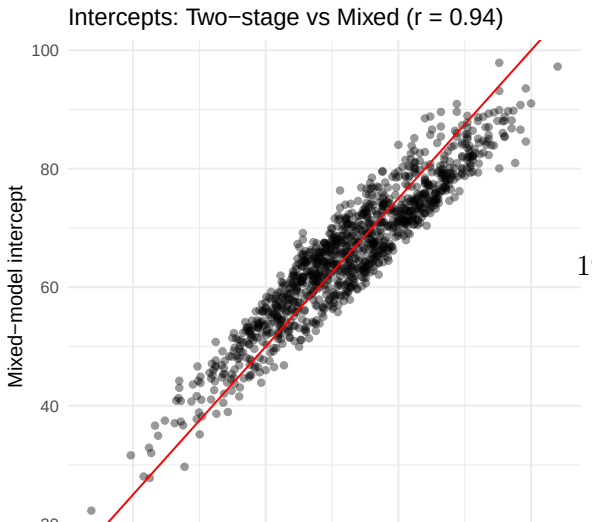
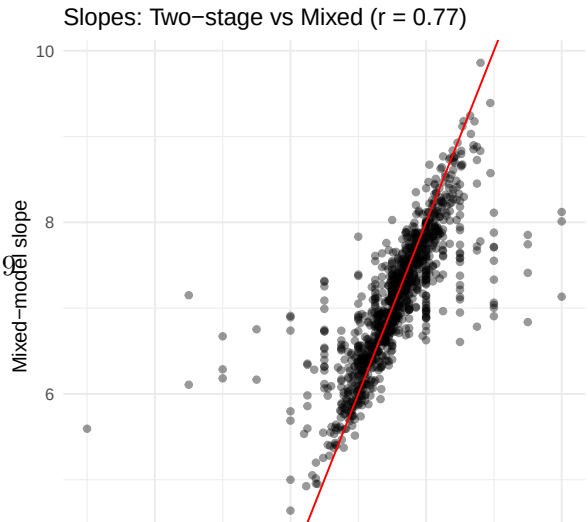
Subject-specific intercepts and slopes from mixed model vs two-stage estimates

Table 10: Two-Stage Slope Model Results

term	estimate	std.error	statistic	p.value
(Intercept)	6.37 [2.97, 9.77]	1.73	3.68	<0.01
sexFemale	-0.04 [-0.19, 0.11]	0.08	-0.55	0.58
age	0.01 [-0.01, 0.04]	0.01	1.02	0.31
bmi	0.00 [-0.03, 0.04]	0.02	0.09	0.93
jobPaid Job	-0.34 [-0.85, 0.17]	0.26	-1.32	0.19
adl	0.05 [-0.02, 0.11]	0.03	1.37	0.17
wzcNursing Home	0.19 [0.02, 0.36]	0.09	2.23	0.03
cdrsb0	-0.02 [-0.03, -0.01]	0.01	-4.30	<0.01
abpet0	-0.07 [-0.34, 0.20]	0.14	-0.53	0.60
taupet0	-0.13 [-1.61, 1.34]	0.75	-0.17	0.86

Table 11: Mixed Model with Random Intercept and Slope Results

effect	term	estimate	std.error	statistic	p.value
fixed	(Intercept)	-60.29 [-68.48, -52.11]	4.17	-14.46	<0.01
fixed	time	7.22 [7.14, 7.30]	0.04	179.32	<0.01
fixed	sexFemale	-0.16 [-0.33, 0.01]	0.09	-1.84	0.07
fixed	age	1.70 [1.62, 1.78]	0.04	42.36	<0.01
fixed	bmi	0.18 [0.12, 0.23]	0.03	6.54	<0.01
fixed	jobPaid Job	-5.14 [-6.71, -3.56]	0.80	-6.39	<0.01
fixed	adl	0.07 [-0.17, 0.30]	0.12	0.55	0.58
fixed	wzcNursing Home	1.92 [1.72, 2.11]	0.10	19.56	<0.01
fixed	cdrsb0	0.00 [-0.01, 0.02]	0.01	0.26	0.79
fixed	time:cdrsb0	-0.02 [-0.03, -0.01]	0.00	-5.25	<0.01

Table 12: Random effects: variance components (lmer)

	Groups	Name	Variance	Std.Dev.	Corr
(Intercept)	patid	(Intercept)	0.807	0.898	NA
time	patid	time	0.686	0.828	-0.611
(Intercept)1	trial	(Intercept)	8.004	2.829	NA
1	Residual		2.698	1.643	NA